Weekly Meeting

25.08.15

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Alpha-value tuning

- Generate w. trained MLP

$$\ell_i = f_{\mathrm{LLM}}(x_{1:t-1})[i], \quad i = 1, \ldots, V$$

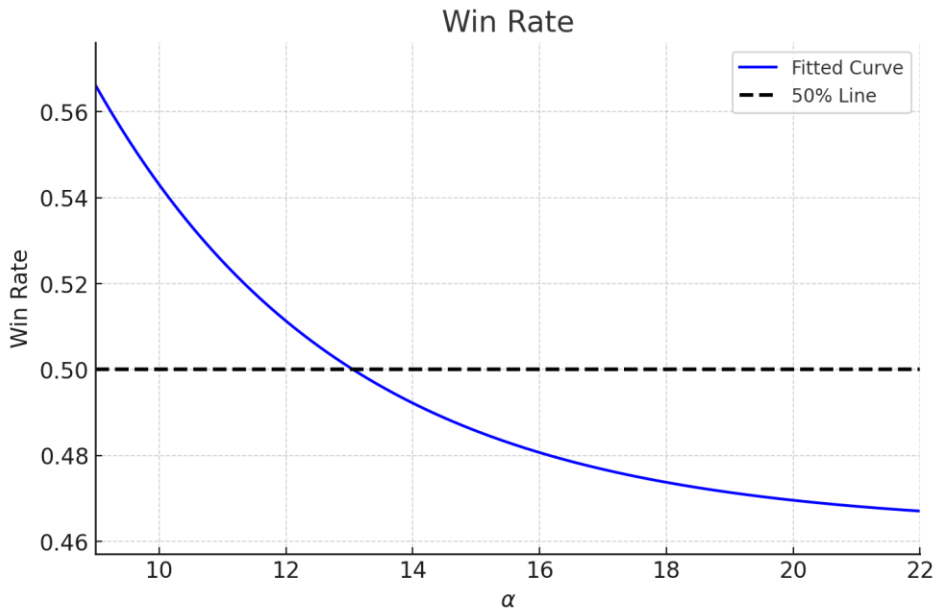
$$\ell'_i = \begin{cases} \ell_i + \alpha\, s_i, & i \in \mathcal{K} \\ \ell_i, & i \notin \mathcal{K} \end{cases}$$

  - s_i: MLP output from last hidden layer & next-token embedding

- Using the original model
  - 8192 → 8 → 1
  - Dataset: alpaca eval

Alpha-value tuning

- Alpha value:
  - $\alpha \approx 1$: No significant difference from baseline
  - $\alpha \geq 10$: Significant change
  - $\alpha \gg 10$: Large α → lower win rates



Dataset Expansion

- Chatbot Arena
  - lmsys/chatbot_arena_conversations
  - Randomly sample 2000 queries
    - (only English, one-turn conversation)

- Other candidates
  - Consider OpenHermes or others
  - Dataset contains a lot of noise
  - Not similar to AlpacaEval dataset

Dataset Expansion

- Chatbot arena:
  - 3498 pair-wise dataset (Exclude cycles. i.e. a>b>c>a)
  - 3298 train / 200 valid / 283 test

- Total (Alpaca eval + Chatbot arena):
  - 4914 pair-wise dataset (2457 instruction)
  - 4156 train / 484 valid / 484 test

Dataset Expansion

- Pair-wise test:
  - Generate: llama3.1, gpt-4o-mini (same setting as before)
  - Llama vs gpt win rate:
    - 587:1412
    - 572:1427
    - Detect preference bias (Include quality bias)
  - Llama vs Llama:
    - 1021:978

MLP train - 1

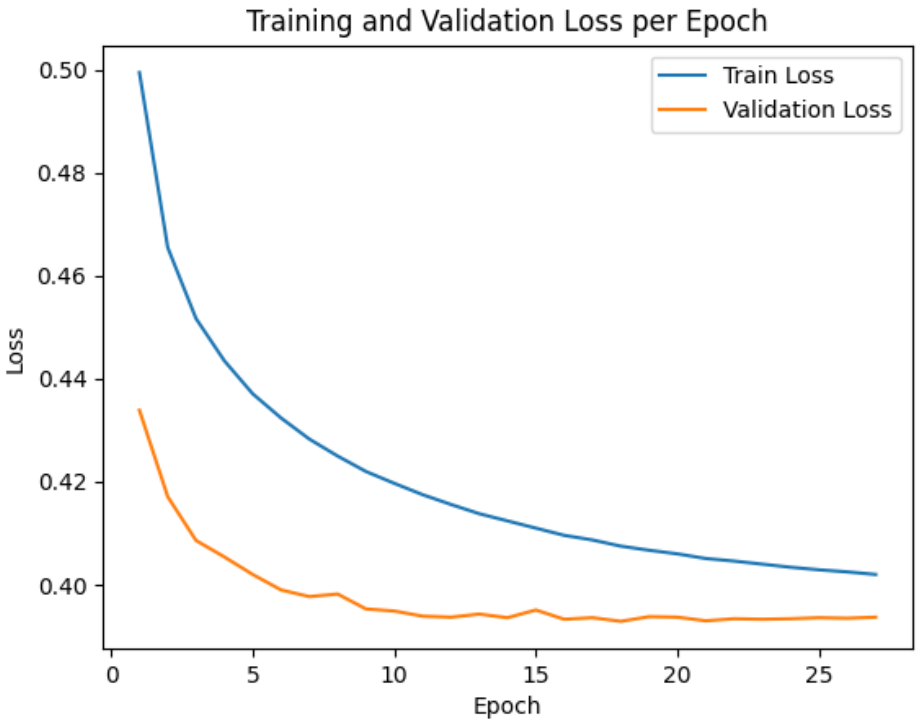
- Hyper-Parameters
  - Optimizer: SGD
  - Momentum: 0.9
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 50
  - Dropout: 0.25
  - Hidden width: 8

- Input dimension: 8192
  - (8192, 8) → GELU → (8, 1)

MLP train - 1

- Results
  - Improvement in accuracy
  - No dramatic performance improvement

```
ep27  train 0.4020 | val 0.3937  acc 64.703%
■ Early stopping triggered
[TEST] loss 0.3885 | acc 63.876%
```



MLP train - 2

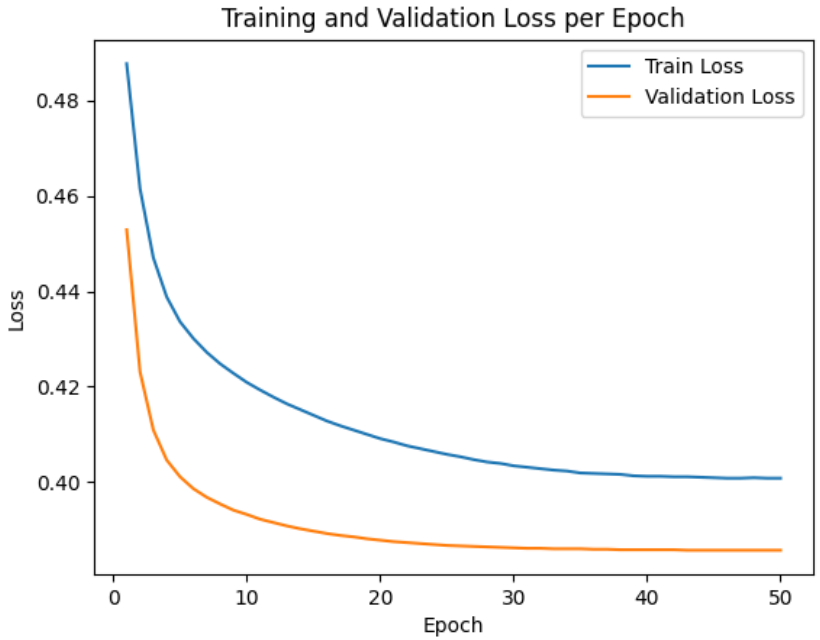
- Hyper-Parameters
  - Optimizer: SGD
  - Momentum: 0.9
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 50
  - Dropout: 0.2
  - Hidden width: 256 - 128

- Input dimension: 8192
  - (8192, 256) → GELU → (256, 128) → GELU → (128, 1)

MLP train - 2

- Results
  - Improvement in accuracy
  - No dramatic performance improvement
  - Parameter ↑ → No change in accuracy

```
ep50  train 0.4008 | val 0.3857  acc 64.926%
  ↳ saved pref_head_reg.pt
[TEST] loss 0.3969 | acc 63.612%
```



Discussion & TODO

- Hyper parameter tuning
  - In progress
  - Increase layers & width
- Num of parameters ↔ accuracy
  - Why performance is similar?
- Num of train datasets ↔ accuracy
  - Training dataset: nearly 4× larger
  - Why performance increases only slightly?